CERTAIN-COAST — Mathematical state of the art

Mathematical state of the art: random forcing, boundaries, initial state and propagation of uncertainty in a stochastic coastal aquifer

NATO SPS Multi-Year Project, call 2027-1. Supplement to the methods of WP2 and WP3. Equations are native Word math; references were checked in Crossref; Tables 2–4 are computed in closed form and checked numerically.

A model that issues warnings for a coastal aquifer must answer four mathematical questions before its probabilities can be trusted. Is the model, driven by random rain, sea level and pumping, well posed? How are the sea and the rivers represented when they are themselves uncertain? How much does the unknown initial state matter? And how are the probabilities of rare events computed at an affordable cost? This note summarises how the literature answers each question, with emphasis on 2023–2026, states what remains open, and gives the rule by which CERTAIN-COAST chooses a method for each question at Fushë-Kuqe (WP2, WP3). The numbers in Tables 2–4 come from closed-form results checked numerically; they are illustrative, not results at the site.

Table 1. The four questions, the mathematical tool that answers each, the number that decides the method, and the point that remains open.

Question	Mathematical tool	Deciding number	Open point
1. Random forcing	Pathwise and entropy solutions; shot-noise law of storm recharge	Π = λτ, storms per aquifer memory	Which law of the forcing to use
2. Boundaries	Conductance (Robin) condition with coloured noise; boundary corrections in the state vector	Γ = CL/T, boundary conductance number	Three-dimensional variable-density aquifer with a moving interface
3. Initial state	Modal memory; ensemble Kalman filter; 4D-Var; Bayesian inversion in function space	t/τ_1, record length over aquifer memory	Assimilation theory for non-additive, non-Gaussian forcing
4. Propagation	Multilevel Monte Carlo with smoothed indicators and surrogate control variates	β − γ, variance decay against cost growth	Rates for threshold probabilities of coupled models

1. The equations

The water table h of an unconfined coastal aquifer obeys the Boussinesq equation, and the unsaturated zone the Richards equation:

$$S_y \frac{\partial h}{\partial t} = \nabla \cdot (K h \, \nabla h) + R(x,t) - Q(x,t) \tag{1}$$

$$\frac{\partial \theta(\psi)}{\partial t} = \nabla \cdot \left(K(\psi) \, \nabla (\psi + z) \right) + R(x,t) \tag{2}$$

Both are degenerate: the diffusion coefficient $K h$ vanishes where the aquifer runs dry, and $K(\psi)$ collapses in dry soil. With $u = h$ and $m = 2$, the Boussinesq equation is a porous-medium equation, and with random forcing it reads

$$du = \Delta(u^m)\,dt + \sigma(u)\,dW_t + R\,dt, \; m > 1 \tag{3}$$

where W is a noise in time: white, with long memory, or rough. The sea enters at the coast and the rivers along their beds; the interface between fresh and salt water is a moving boundary.

2. Random forcing

2.1 Existence, uniqueness and stability

For noise that is additive or linear in u, the substitution $u = e^{\mu W_t} v$ turns the stochastic equation into a porous-medium equation with random coefficients, which can be solved path by path. Pathwise well-posedness is proved for stochastic porous-medium and fast-diffusion equations with conservative noise (Fehrman and Gess, 2019), and on bounded domains with Dirichlet data, where the solution depends continuously on the driving noise (Clini, 2023). For nonlinear noise that is correlated in space, stochastic kinetic solutions give well-posedness (Fehrman and Gess, 2024; Popat, 2025). For a given path of the forcing, the porous-medium operator is accretive in L^1, so two solutions driven by the same forcing contract (Vázquez, 2006):

$$\left\| u_1(t) - u_2(t) \right\|_{L^1} \le \left\| u_1(0) - u_2(0) \right\|_{L^1}, \tag{4}$$

Together with the continuity in the driving noise, this means that an error in the forcing has a controlled effect on the heads. For forcing rougher than Brownian motion, Wong–Zakai regularisation and splitting give fully discrete schemes with error bounds (He, 2023; Nie et al., 2025).

2.2 Heads that stay nonnegative

A finite-volume discretisation with two-point fluxes and implicit time steps gives an M-matrix, and the discrete maximum principle then keeps $h \ge 0$ in cells that dry out. For multiplicative noise, splitting the noise step from the diffusion step and solving the noise step exactly preserves positivity (Bréhier et al., 2024); finite-element schemes that keep the stochastic porous-medium equation nonnegative converge (Grillmeier and Grün, 2019), and convergent schemes now cover singular-degenerate stochastic equations that include it (Baňas et al., 2024).

2.3 The law of the forcing

What the theory does not decide is the law of the forcing. Rain falls as storms: at the scale of an aquifer, recharge is a marked Poisson process with storm rate λ and storm sizes Y_i of mean μ, written as the rise in head they cause. If the aquifer drains like a linear reservoir with response time τ, the head above its drainage level is shot noise:

$$h(t) = \sum_{t_i \le t} Y_i \, e^{-(t - t_i)/\tau} \tag{5}$$

For exponential storm sizes its stationary law is exactly a Gamma law, with shape $\Pi = \lambda \tau$, the number of storms within one aquifer memory, and scale μ (the Laplace transform of (5) is $(1 + s\mu)^{-\lambda\tau}$; the same law describes soil moisture in Rodriguez-Iturbe et al., 1999; for the water table, Laio et al., 2009):

$$h \sim \mathrm{Gamma}(\Pi, \mu), \; E[h] = \Pi \mu, \; \mathrm{Var}[h] = \Pi \mu^2, \; \mathrm{skewness} = 2/\sqrt{\Pi} \tag{6}$$

A Gaussian with the same mean and variance is exact only as $\Pi \to \infty$. Table 2 shows how far it is from the exact law in the tails that warnings use. The Gaussian underestimates high levels by an order of magnitude when few storms fall within one aquifer memory, and still by a factor of two at $\Pi = 100$; it overstates low levels, which matter for pumping limits. Means and variances are right at any Π; tails are not. A search of the 2023–2026 literature for this note found no study that compares the two laws in the tails of groundwater levels.

Table 2. Tail probabilities of the head under the exact Gamma law and under a Gaussian with the same mean and variance (sd: standard deviation).

Π = λτ	P[h > mean + 3 sd], exact	Gaussian	Ratio	P[h < mean − 2 sd], exact	Gaussian
1	1.83%	0.135%	13.6	0	2.28%
2	1.41%	0.135%	10.4	0	2.28%
10	0.67%	0.135%	5.0	0.46%	2.28%
100	0.28%	0.135%	2.0	1.71%	2.28%

3. Boundaries

3.1 The sea and the rivers as conductance boundaries

The sea at the coast and the rivers along their beds are written as a Robin (conductance) condition,

$$-T\frac{\partial h}{\partial n}=C\left(h-H_b(t)\right),H_b(t)=\acute{H}+\delta+\xi(t), \tag{7}$$

with transmissivity T, boundary conductance C, mean stage $\acute{H}$, an unknown datum correction δ and a fluctuation ξ. The problem is well posed for $C\geq 0$, because the bilinear form is coercive (Lax–Milgram), and the number $\Gamma = C\,L/T$ decides how the boundary acts: for $\Gamma \to \infty$ it is a fixed head, for $\Gamma \to 0$ a no-flow boundary. A stochastic porous-medium equation with Robin boundary conditions, gravity-driven infiltration and multiplicative noise is treated in Ciotir et al. (2025), and a rule for when boundary uncertainty must enter the estimation of aquifer properties is given in Peeters and Turnadge (2019). For a strip of length L with a conductance boundary at the coast and a divide inland, the slowest decay rate solves $k_1 L\tan\left(k_1 L\right)=\Gamma$, and the aquifer memory is $\tau_1=S/\left(T\,k_1^2\right)$. Table 3 shows that a sea or river boundary behaves as a fixed head within 2% only for $\Gamma\geq 100$.

Table 3. Memory of the aquifer with a conductance boundary, relative to a fixed head at the coast.

Γ = CL/T	k_1L	τ_1 relative to a fixed head
0.1	0.311	25.5
1	0.860	3.33
10	1.429	1.21
100	1.555	1.02
∞	π/2	1

White noise imposed directly as fixed-head data is too rough for classical solutions. The fluctuation of the sea is therefore taken as a coloured (Ornstein–Uhlenbeck) process, with correlation time τ_ξ and intensity σ_ξ estimated from the tide gauge:

$$d\xi=-\frac{\xi}{\tau_\xi}dt+\sigma_\xi dW_t \tag{8}$$

3.2 The interface between fresh and salt water

In the sharp-interface approximation, the depth of the interface below sea level is $\zeta=\rho_f h/\left(\rho_s-\rho_f\right)\approx 40\,h$ (Ghyben–Herzberg), and a single potential turns the regional interface problem into one equation over the whole aquifer (Strack, 1976). With random fluxes the interface is a stochastic free boundary of Stefan type. Such problems have local solutions after a transformation to a fixed domain (Müller, 2018), and maximal solutions up to stopping times (Antonopoulou et al., 2025); none of them yet covers a variable-density aquifer.

3.3 A wrong boundary is read as wrong recharge

Write the heads as $y=G_R r+G_B b+e$, with r the recharge and b the boundary data. If b is fixed with an error δb, the least-squares estimate of the recharge is shifted by

$$E\left[\hat{r}\right]-r=\left(G_R^T G_R\right)^{-1}G_R^T G_B\,\delta b, \tag{9}$$

so a boundary error is read as a recharge error; a related non-uniqueness lets a wrong recharge field fit the heads (Erdal and Cirpka, 2016). The remedy is to put the datum and stage corrections into the state vector, $z=\left(h,r,\delta_{\text{sea}},\delta_{river}\right)$, and estimate them with the heads, as ensemble history matching already does for sea-level and recharge ensembles of an island aquifer (Coulon et al., 2024). The boundary and recharge uncertainty that remains then sets the floor of the predictive uncertainty, which more head data alone do not remove.

4. Initial state

4.1 How long the aquifer remembers

In the linearised aquifer, the head is a sum of decaying modes plus the response to the forcing:

$$h(x,t)=\sum_{k\geq 1} a_k e^{-t/\tau_k}\varphi_k(x)+h_f(x,t),\tau_1=\frac{4SL^2}{\pi^2 T}, \tag{10}$$

for a fixed head at the coast and a divide at distance L. An error in the initial state decays as e^{-t/τ_1}, so a spin-up of $\tau_1\ln 10\approx 2.3\,\tau_1$ leaves 10% of it. The same time scale limits what data can say about the initial state: with observation noise of intensity σ^2, the Fisher information on a_1 from data after time t is

$$I_{a_1}(t)=\frac{1}{\sigma^2}\int_t^\infty e^{-2s/\tau_1}ds=\frac{\tau_1}{2\sigma^2}e^{-2t/\tau_1}, \tag{11}$$

so the initial state is forgotten, and becomes unidentifiable, on the same time scale. Either the model is spun up for $2.3\,\tau_1$, or the initial heads are estimated from the first part of the record, shorter than τ_1.

4.2 Ensemble Kalman filtering

The ensemble Kalman filter (Evensen, 1994) updates each member of the ensemble with

$$x^a=x^f+K\left(y-Hx^f\right),K=P^fH^\top\left(HP^fH^\top+R\right)^{-1}. \tag{12}$$

It is exact in the linear-Gaussian case as the ensemble grows, and recent mean-field analysis bounds its error when the forecast is close to Gaussian or the dynamics close to linear (Carrillo et al., 2024; Calvello et al., 2026), within a mean-field framework for ensemble Kalman methods (Calvello et al., 2025); non-asymptotic analysis explains when a small ensemble suffices (Al-Ghattas and Sanz-Alonso, 2024). In practice its accuracy is monitored through the skewness and kurtosis of the forecast ensemble; transport-map filters extend it when the forecast is far from Gaussian (Spantini et al., 2022).

4.3 Variational estimation

4D-Var estimates the initial state by minimising

$$J(x_0)=1/2\left\|x_0-x_b\right\|^2_{B^{-1}}+1/2\sum_k\left\|y_k-H\,M_k(x_0)\right\|^2_{R^{-1}}, \tag{13}$$

with the gradient from the adjoint model. For a nonlinear aquifer model, the Gauss–Newton iteration needs safeguards: plain Gauss–Newton may fail to converge from a poor starting point, line-search and regularised variants restore convergence (Cartis et al., 2024), and Levenberg–Marquardt methods with inexact steps converge globally (Bergou et al., 2016).

4.4 Bayesian inversion in function space

Bayesian inversion in function space (Stuart, 2010) gives posteriors for initial heads and parameters that do not degrade as the mesh is refined; Laplace approximations with low-rank or hierarchical Hessians make them scalable (Hartland et al., 2023; Kim et al., 2023), and goal-oriented real-time Bayesian inference now serves early warning, for tsunamis (Henneking et al., 2026).

5. Propagation of uncertainty

5.1 Multilevel Monte Carlo

The probabilities that warnings need, of heads below a pumping threshold or of chloride above a limit, are expectations of functionals P of the solution. Multilevel Monte Carlo writes the expectation on the finest mesh as a telescoping sum over meshes $l=0,\dots,L$ (Giles, 2008):

$$E\left[P_L\right]=E\left[P_0\right]+\sum_{l=1}^{L}E\left[P_l-P_{l-1}\right]. \tag{14}$$

If the weak error decays as $2^{-\alpha l}$, the variance of the corrections as $2^{-\beta l}$ and the cost of one sample grows as $2^{\gamma l}$, the cost of a root-mean-square accuracy ε is (Giles, 2015)

$$\mathrm{Cost}(\varepsilon) \propto \begin{cases} \varepsilon^{-2}, & \beta > \gamma, \\ \varepsilon^{-2}(\log \varepsilon)^2, & \beta = \gamma, \\ \varepsilon^{-2-(\gamma-\beta)/\alpha}, & \beta < \gamma, \end{cases} \tag{15}$$

so when the corrections shrink faster than the cost grows, a probability on the finest mesh costs as much, in order, as an average of independent numbers (for groundwater flow with random coefficients, Istratuca and Teckentrup, 2025).

5.2 Probabilities of crossing a threshold

For a probability the functional is an indicator, $P = 1\{h < h_{min}\}$, which is discontinuous and lowers β. Smoothing the indicator, by a conditional expectation over the last step or by a kernel, restores the rate (Haji-Ali et al., 2022; Bayer et al., 2024). Surrogates enter as control variates on the coarse levels (Peherstorfer et al., 2016; Gruber et al., 2023).

5.3 Model error apart from forcing

Model error is kept apart from the forcing by writing $x_{k+1} = M(x_k, r_k) + \eta_k$, with η_k estimated apart from the recharge r_k; otherwise the filter attributes model bias to the rain. Weak-constraint 4D-Var estimates η_k as part of the control variable (Trémolet, 2007), and its covariance can be chosen as a regularisation parameter (Babyale et al., 2025); but model error and initial state separate only under specific conditions (Laloyaux et al., 2020), and in hydrology rainfall and structural errors can be poorly identifiable (Renard et al., 2010).

5.4 Sampling and averaging

Averaging a process with correlation time τ over intervals of length Δ changes its correlation. For an Ornstein–Uhlenbeck process with variance σ^2, the block means satisfy

$$\mathrm{Var}(\acute{A}) = 2\sigma^2 \frac{\tau}{\Delta}\left[1 - \frac{\tau}{\Delta}\left(1 - e^{-\Delta/\tau}\right)\right], \mathrm{Cov}(\acute{A}_k, \acute{A}_{k+1}) = \sigma^2 \frac{\tau^2}{\Delta^2}\left(1 - e^{-\Delta/\tau}\right)^2, \tag{16}$$

so the lag-one correlation of monthly means is not $e^{-\Delta/\tau}$, and a response time read from monthly means is biased upwards (Table 4). The bias of estimators for a time-integrated Ornstein–Uhlenbeck process is derived in econometrics (Hoyos, 2025), and averaging recessions is known to bias recession parameters in hydrology (Jachens et al., 2020), but we found no study that quantifies the effect on groundwater response times. Daily logger data avoid the bias.

Table 4. Response time of the aquifer estimated from monthly means (Δ = 30 days) against the true value.

True response time τ (days)	Estimated from monthly means (days)	Overestimate
2	9.0	4.5 ×
5	13.0	2.6 ×
10	19.8	2.0 ×
30	49.1	1.6 ×
90	138.9	1.5 ×

6. What is new: four numbers that choose the method

The literature answers each question separately. CERTAIN-COAST joins the answers through four dimensionless numbers computed from the records of Fushë-Kuqe before any simulation (Table 5). Each number selects the method for one question, so the workflow is fixed by data rather than by habit, and each approximation comes with a stated error: the exact Gamma law of Section 2.3 is the reference against which the Gaussian forcing is checked (Table 2), and the averaging bias of Section 5.4 is the reason the network of WP4 records daily (Table 4). The individual results are known; their use as one decision map for a coastal aquifer is the contribution.

Table 5. The decision map: four numbers, computed from data, that choose the method for each question.

Number	Computed from	Rule and method chosen
$\Pi = \lambda\tau$	Storm rate from the rain gauge; response time from head recessions	$\Pi \geq 20$ and only means and variances needed: Gaussian recharge noise. Otherwise shot-noise forcing; tail probabilities always from the exact Gamma law.
$\Gamma = CL/T$	Conductance of the river beds and of the coast; transmissivity	$\Gamma \geq 100$: fixed head with an uncertain datum. $\Gamma \leq 0.1$: no flow. Between: conductance condition with the stage correction in the state vector.
t/τ_1	Storage, transmissivity and length of the model; length of the record	Record before the forecast $\geq 2.3\ \tau_1$: spin-up. Otherwise: initial heads estimated from the first part of the record.
$\beta - \gamma$	Pilot runs on three meshes	$\beta > \gamma$: standard multilevel Monte Carlo. $\beta \leq \gamma$: smoothed indicators and surrogate control variates.

7. What remains open

No well-posedness theorem covers a three-dimensional, variable-density coastal aquifer with random recharge, random sea level and a moving interface at once; most results are one-dimensional, periodic or with homogeneous Dirichlet conditions. Rainfall generators with temporal clustering have no theory for the aquifer response beyond the shot-noise case. The theorems on assimilation assume bounded dynamics and additive Gaussian forcing, so the multiplicative, heavy-tailed forcing of Section 2 lies outside them. CERTAIN-COAST does not claim to close these gaps: it uses the decision map to stay where the theory holds and reports where it does not.

8. Where this enters the work plan

T2.3, calibration: the law of the recharge chosen by Π; datum and stage corrections in the state vector where Γ requires it; initial heads by spin-up or estimation according to t/τ_1. **T3.2, alarms:** tail probabilities from the exact law and multilevel Monte Carlo with smoothed indicators. **T4.1, network design:** daily loggers (Table 4), a local rain gauge for the storm rate, and the tide gauge for the fluctuation of the sea. **WP5, live operation:** the non-Gaussianity of the forecast ensemble is monitored at every update.

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